high fees underperformed those with low fees, and both underperformed index funds having the lowest fees.

In a similar study, Barras, Scaillet, and Wermers (2010) studied 2,076 U.S. mutual funds from 1989 through 2006. Adjusting for data snooping bias, they concluded that the number of funds with skilled managers where active returns exceeded costs was statistically indistinguishable from zero.[17] They also found that the fraction of skilled managers declined from 14.40% to 0.60% from 1989 to 2006. The authors attributed this shift in performance to an increase in unskilled managers who nonetheless charged high fees with funds having high expense ratios.

According to Morningstar, the average annual expense ratio of an actively managed mutual fund is 1.41%, compared to 0.20% for a passively managed fund. Actively managed funds also have, on average, a portfolio turnover ratio of 83%, which adds an additional 0.70% per year in transactions cost expense. Furthermore, tax inefficiencies can add an additional 1% per year to the costs of owning an actively managed fund. Not accounting for possible negative tax consequences, Morningstar reported that at the end of 2012, after adjusting for survivorship bias, over 80% of large-blend mutual funds underperformed their benchmarks over the past 3, 5, 10, and 15 years.

Credit Suisse reported that the annual total return of the S&P 500 Index for the last 20 years ending in 2013 was 9.3%. The average actively managed mutual fund during this period earned 1.0% to 1.5% less than that, due to its expense ratios and transaction costs. The average returns investors earned was another 1% to 2% less than that, due to poor timing decisions. The overall return for investors in actively managed funds was 60% to 80% less than the market return over the past 20 years.[18] According to John Bogle, founder and former CEO of The Vanguard Group (now the world's largest mutual fund company), "Selecting funds that will significantly exceed market returns, a search in which hope springs eternal and in which past performance has proven of virtually no predictive value, is a loser's game."

Vanguard has an interactive website where you can input an active fund's expense ratio and see how your capital would grow at a 6% growth rate, compared to an index fund with an expense ratio of 0.25%.[19] If we input the average annual active fund expense ratio of 1.41%, at the end of 50 years we